

CDACL-002

Customer Segmentation Analysis

Client Project Report

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Datamites – Certified Data Analyst (CDA) Program

Tools: Power BI · Python (KMeans Clustering) · MySQL

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1. Project Introduction

This project analyzes a US retail shopping-trends dataset of 7,800 customer transactions to uncover purchasing patterns and support data-driven marketing decisions. The analysis addresses three business questions defined in the project brief:

- Which product categories should receive discounts, and why.
- How customer age, along with season and location, affects card spending.
- How customers can be segmented into groups (via clustering) to guide targeted marketing campaigns.

The analysis combines Python (KMeans clustering) for customer segmentation with an interactive, 4-page Power BI dashboard for exploration and reporting.

2. Dataset Overview

Attribute	Details
Source	MySQL database (project_shopping_trends)
Records	7,800 US shoppers
Key fields	Customer ID, Age, Gender, Category, Purchase Amount, Location, Season, Review Rating, Subscription Status, Discount Applied, Promo Code Used, Previous Purchases, Payment Method, Purchase Frequency

3. Methodology

- Data cleaned and de-duplicated using Power Query (MySQL source).
- Customer segments generated using Python KMeans clustering on spending behavior.
- Cluster output merged back with the main dataset via Customer ID.
- Interactive 4-page Power BI dashboard built for Overview, Discount Analysis, Spending Analysis, and Segmentation.

4. Dashboard Overview

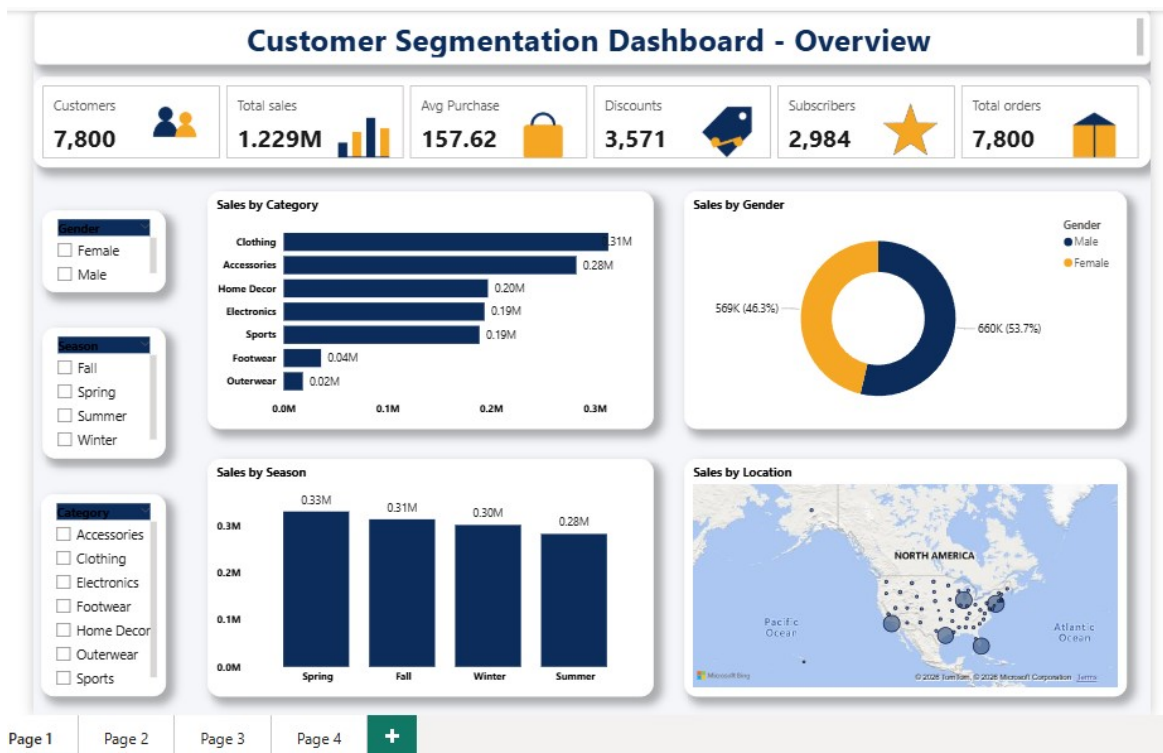


Figure 1: Customer Segmentation Dashboard – Overview

The overview page summarizes overall performance across 7,800 order records, totaling approximately \$1.229M in sales with an average purchase value of \$157.62.

Metric	Value	Metric	Value
Customers	7,800	Discounts	3,571
Total Sales	1.229M	Subscribers	2,984
Avg Purchase	157.62	Total Orders	7,800

- Clothing is the top-selling category (0.31M), followed by Accessories (0.28M); Outerwear is lowest (0.02M).
- Gender split is fairly balanced: Male 53.7%, Female 46.3%.
- Spring is the strongest season for sales (0.33M), tapering through Summer (0.28M).
- Sales are concentrated across North American locations, as shown on the map.

5. Discount Analysis — Question 1



Figure 2: Discount Analysis Dashboard

Key Observations

- Clothing and Accessories receive the highest discount counts (1,102 and 950) — aligning with their position as the top two categories by sales volume.
- Footwear (3.79) and Outerwear (3.75) have the highest average ratings, despite receiving the fewest discounts (259 and 144) and lowest sales.
- Home Decor (2.98) and Electronics (3.06) have the lowest ratings despite moderate discount activity, suggesting discounting alone isn't resolving satisfaction gaps.
- Average previous purchases are broadly similar across categories (~25 units), showing consistent repeat-purchase behavior regardless of category.

Recommendation

Discounts on Clothing and Accessories are working well and driving volume — continue the current strategy. Footwear and Outerwear are highly rated but under-discounted and under-sold; targeted promotions here could unlock demand from an already satisfied customer base. For Home Decor and Electronics, discounting should be paired with quality or service improvements, since ratings remain low despite existing discount activity.

6. Age, Season & Location Spending — Question 2

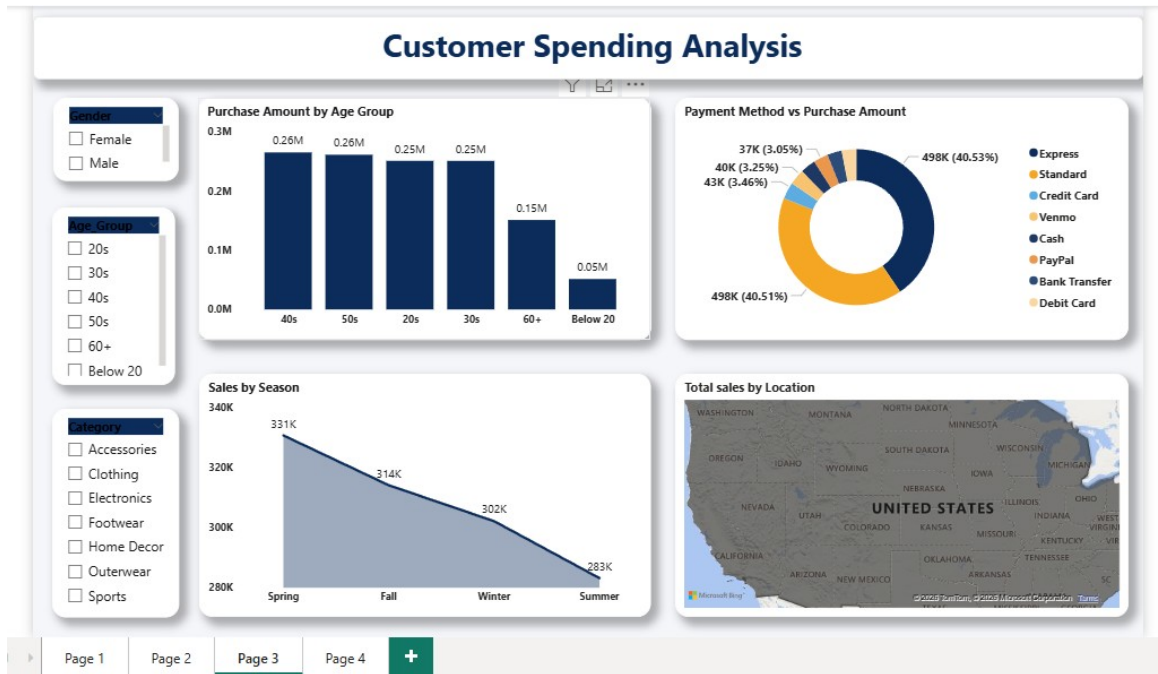


Figure 3: Customer Spending Analysis Dashboard

Key Observations

- Spending is concentrated in the 20s–50s age groups (0.25M–0.26M each); Below-20 (0.05M) and 60+ (0.15M) contribute far less.
- Express and Standard methods dominate transaction value, each ~40% of purchase amount; all other methods are minor by comparison.
- Seasonally, spending peaks in Spring (331K) and declines through Fall (314K), Winter (302K), to Summer (283K) — an 18% drop from peak to lowest.
- Purchases are spread nationally across the United States with no single dominant state.

Recommendation

Marketing spend should prioritize the 20–50 age bracket, which drives the bulk of revenue, while lightweight campaigns (referral or student offers) could grow the under-20 segment. Since Spring generates the highest spend, pre-season promotions ahead of Spring and re-engagement campaigns in Summer (the weakest season) could help flatten the seasonal dip.

7. Customer Segmentation — Question 3

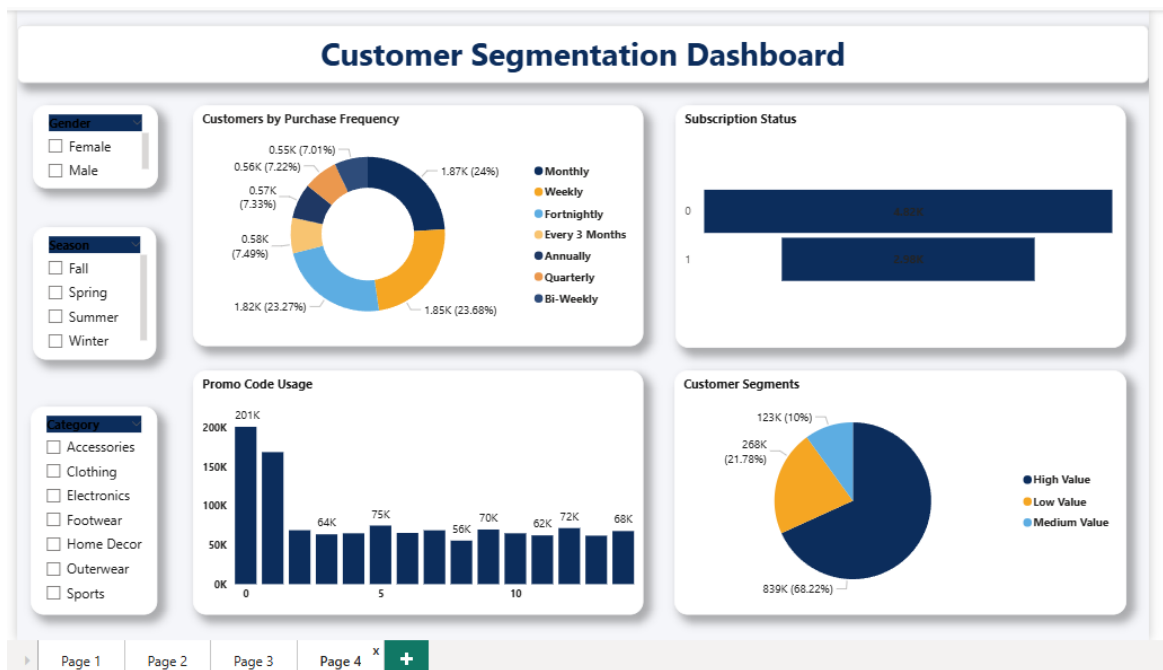


Figure 4: Customer Segmentation Dashboard

Value Segments

Segment	Customers	% of Base
High Value	839K	68.2%
Medium Value	123K	10.0%
Low Value	268K	21.8%

Key Observations

- The majority of customers (68.2%) fall into the High Value segment — a strong core customer base.
- Purchase frequency is evenly spread across Monthly (24%), Weekly (23.7%), and Fortnightly (23.3%) buyers, with smaller shares for the remaining cadences.
- Promo code usage is concentrated at a single-use level (201K), dropping sharply for repeat use — most customers redeem a code once rather than repeatedly.

Marketing Recommendations by Segment

- High Value: Focus on retention — loyalty programs, early access to new arrivals, personalized offers.
- Medium Value: Targeted upsell campaigns (bundles, cross-sell) to migrate them toward High Value.
- Low Value: Re-engagement campaigns with entry-level discounts or free shipping to lift purchase frequency.

8. Overall Conclusion

The analysis shows a fashion-led retail business where Clothing and Accessories dominate both sales and discount activity, core middle-aged customers (20s–50s) drive the majority of revenue, and a strong majority of customers (68%) are already high-value. The clearest opportunities are:

- Extending targeted discounts to high-rated, under-performing categories like Footwear and Outerwear.
- Countering the Spring-to-Summer seasonal decline with timed promotions.
- Building segment-specific marketing — retention for High Value customers and activation offers for Medium/Low Value customers.